

Third Lecture

➤ Graphical presentation for quantitative data:

1- Frequency histogram: A graph that display the classes on the horizontal axis and the frequencies (f_i) on the vertical axis. The frequency of each class is represented by vertical bar.

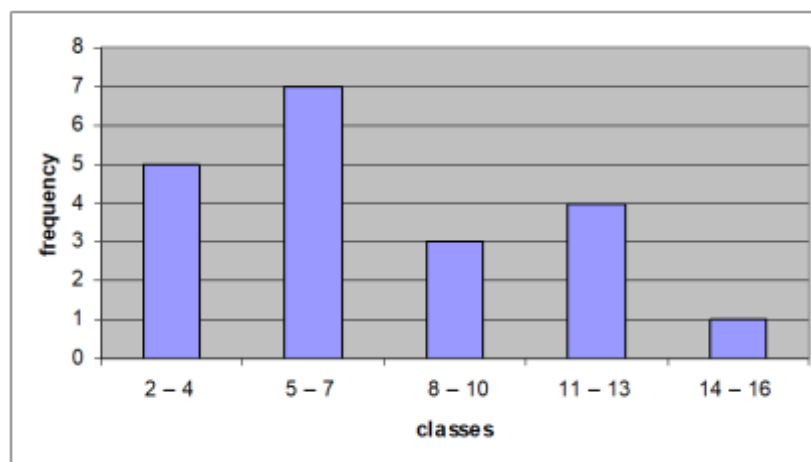
Example: Presenting this data in frequency histogram

□ العرض البياني للبيانات الكمية:

1- المدرج التكراري: رسم بياني يعرض الفئات على المحور الأفقي والتكرارات (f_i) على المحور الرأسي. يُمثل تكرار كل فئة بخط عمودي.

مثال: عرض هذه البيانات في مدرج تكراري

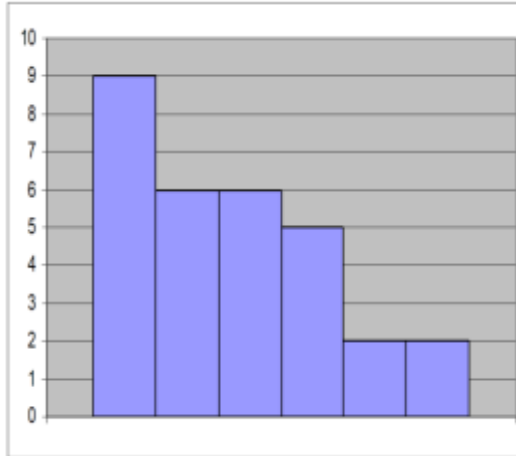
Classes	frequency
2 – 4	5
5 – 7	7
8 – 10	3
11 – 13	4
14 – 16	1



Example: Present this data in frequency histogram

مثال: اعرض هذه البيانات في رسم بياني تكراري

Classes	frequency
5 –	9
9 –	6
13 –	6
17 –	5
21 –	2
25 – 29	2



Example : Draw a histogram to illustrate the following frequency distribution.

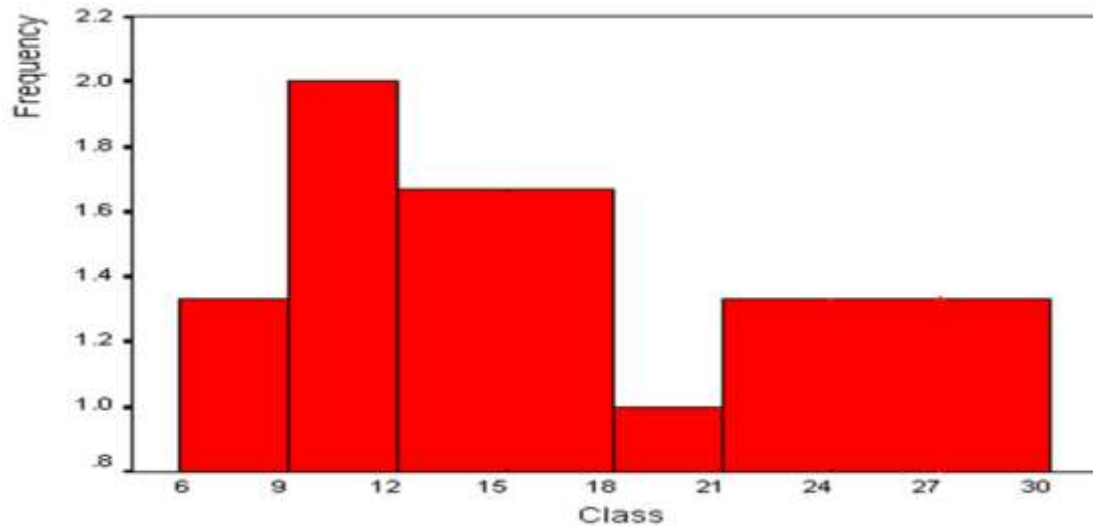
مثال: ارسم مدرجًا تكراريًا لتوضيح توزيع التكرار التالي

Class	Frequency
6 –	4
9 –	6
12 –	10
18 –	3
21 – 30	12
Total	35

Solution: As the class widths are not equal, we should find the adjusted frequency and instead of frequency, we will take it in the consideration

الحل: بما أن عرض الفئات غير متساوي، يجب علينا إيجاد التكرار المعدل، وبدلاً من التكرار، سنأخذه في الاعتبار.

Class	Frequency	Class width	Adjusted frequency
6 –	4	3	$4/3=1.33$
9 –	6	3	$6/3=2$
12 –	10	6	$10/6=1.67$
18 –	3	3	$3/3=1$
21 – 30	12	9	$12/9=1.33$
Total	35		



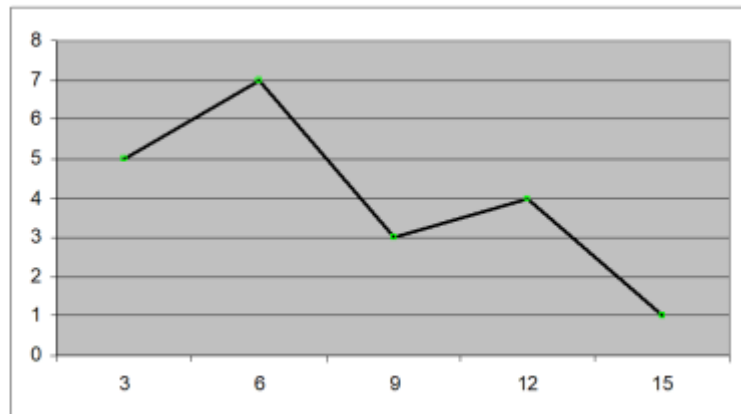
2- Frequency polygon: A graph that display the mid points of classes (X_i) on the horizontal axis and the frequencies (f_i) on the vertical axis. The points are connected by lines.

2- المضلع التكراري: رسم بياني يعرض نقاط منتصف الفئات (X_i) على المحور الأفقي والتكرارات (f_i) على المحور الرأسي. يتم توصيل النقاط بخطوط.

Example: Draw a Frequency polygon to illustrate the following frequency distribution

مثال: ارسم مضلعًا تكراريًا لتوضيح التوزيع التكراري التالي

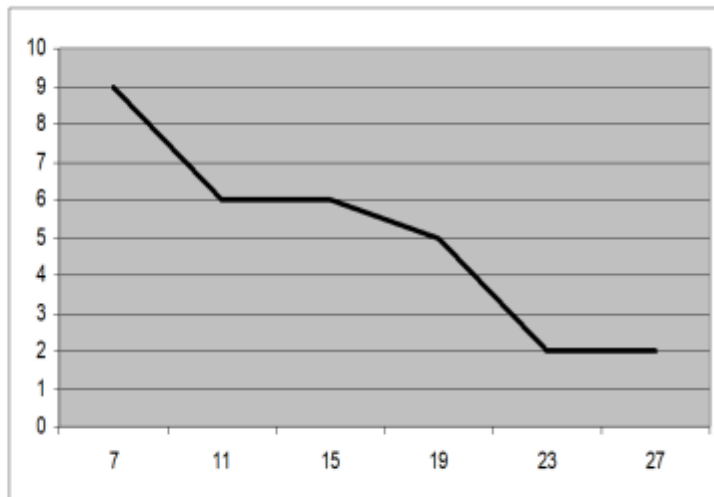
Classes	f_i	X_i
2 – 4	5	3
5 – 7	7	6
8 – 10	3	9
11 – 13	4	12
14 – 16	1	15



Example: Present this data in frequency polygon:

مثال: اعرض هذه البيانات في شكل مضلع تكراري

Classes	f_i	X_i
5 –	9	7
9 –	6	11
13 –	6	15
17 –	5	19
21 –	2	23
25 – 29	2	27



Chapter Three: Notation مجموعة رموز

1. $\sum$ - Notation

Some uses for this symbol بعض استخدامات هذا الرمز

$$1. \sum_{i=1}^n X_i = X_1 + X_2 + \dots + X_n$$

$$2. \sum_{i=1}^n X_i^2 = X_1^2 + X_2^2 + \dots + X_n^2$$

$$3. \left(\sum_{i=1}^n X_i \right)^2 = (X_1 + X_2 + \dots + X_n)^2$$

$$\text{Note : } \sum_{i=1}^n X_i^2 \neq \left(\sum_{i=1}^n X_i \right)^2$$

$$4. \sum_{i=1}^n \frac{1}{X_i} = \frac{1}{X_1} + \frac{1}{X_2} + \dots + \frac{1}{X_n}$$

$$5. \frac{1}{\sum_{i=1}^n X_i} = \frac{1}{X_1 + X_2 + \dots + X_n}$$

$$\text{Note : } \sum_{i=1}^n \frac{1}{X_i} \neq \frac{1}{\sum_{i=1}^n X_i}$$

The properties for this symbol: خصائص هذا الرمز

Let a and b are two constants

ليكن a و b ثابتين

$$1. \sum_{i=1}^n a = a \sum_{i=1}^n 1 = na$$

$$2. \sum_{i=1}^n aX_i = a \sum_{i=1}^n X_i$$

$$3. \sum_{i=1}^n (X_i + a) = \sum_{i=1}^n X_i + na$$

$$4. \sum_{i=1}^n (X_i + Y_i) = \sum_{i=1}^n X_i + \sum_{i=1}^n Y_i$$

$$5. \sum_{i=1}^n (aX_i + bY_i) = a \sum_{i=1}^n X_i + b \sum_{i=1}^n Y_i$$

$$6. \sum_{i=1}^n X_i Y_i = X_1 Y_1 + X_2 Y_2 + \dots + X_n Y_n$$

Example: X_i : 2, 4, 5, 8, 3

Y_i : 1, 2, 3, 2, 3

$$1. \quad \sum_{i=1}^n X_i = \sum_{i=1}^5 X_i = 2 + 4 + 5 + 8 + 3 = 22$$

$$2. \quad \sum_{i=1}^n X_i^2 = \sum_{i=1}^5 X_i^2 = 2^2 + 4^2 + 5^2 + 8^2 + 3^2 = 118$$

$$3. \quad \left(\sum_{i=1}^n X_i \right)^2 = \left(\sum_{i=1}^5 X_i \right)^2 = (2 + 4 + 5 + 8 + 3)^2 = (22)^2 = 484$$

$$4. \quad \sum_{i=1}^3 X_i = 2 + 4 + 5 = 11$$

$$5. \quad \sum_{i=3}^5 X_i = 5 + 8 + 3 = 16$$

$$6. \quad \frac{1}{\sum_{i=1}^n X_i} = \frac{1}{\sum_{i=1}^5 X_i} = \frac{1}{2 + 4 + 5 + 8 + 3} = \frac{1}{22}$$

$$7. \quad \sum_{i=1}^n \frac{1}{X_i} = \sum_{i=1}^5 \frac{1}{X_i} = \frac{1}{2} + \frac{1}{4} + \frac{1}{5} + \frac{1}{8} + \frac{1}{3} = \frac{169}{120}$$

8. $\sum_{i=1}^n 7 = \sum_{i=1}^5 7 = (n)(7) = (5)(7) = 35$
9. $\sum_{i=1}^3 7 = (3)(7) = 21$
10. $\sum_{i=1}^5 3X_i = 3\sum_{i=1}^5 X_i = (3)(22) = 66$
11. $\sum_{i=1}^5 (X_i - 4) = \sum_{i=1}^5 X_i - \sum_{i=1}^5 4 = 22 - (5)(4) = 22 - 20 = 2$
12. $\sum_{i=1}^5 (X_i - Y_i) = \sum_{i=1}^5 X_i - \sum_{i=1}^5 Y_i = 22 - 11 = 11$
13. $\sum_{i=1}^5 X_i Y_i = (2)(1) + (4)(2) + (5)(3) + (8)(2) + (3)(3) = 50$
14. $\sum_{i=1}^5 (X_i - 4)(Y_i - 3) = \sum_{i=1}^5 X_i Y_i - 3\sum_{i=1}^5 X_i - 4\sum_{i=1}^5 Y_i + 5(12) = 50 - 3(22) - 4(11) + 12 =$
15. $\sum_{i=1}^5 (Y_i - 4)^2 = \sum_{i=1}^5 (Y_i^2 - 8Y_i + 16) = \sum_{i=1}^5 Y_i^2 - 8\sum_{i=1}^5 Y_i + (5)(16) = 27 - (8)(11) + 80 = 19$

2. π - Notation

Some uses and properties for this symbol

1. $\prod_{i=1}^n a = a^n$
2. $\prod_{i=1}^n X_i = X_1 \cdot X_2 \cdot X_3 \cdot \dots \cdot X_n$
3. $\prod_{i=1}^n a X_i = a^n \prod_{i=1}^n X_i$
4. $\prod_{i=1}^n X_i Y_i = \prod_{i=1}^n X_i \prod_{i=1}^n Y_i$
5. $\prod_{i=1}^n \frac{X_i}{Y_i} = \frac{\prod_{i=1}^n X_i}{\prod_{i=1}^n Y_i}$
6. $\prod_{i=1}^n X_i^a = \left(\prod_{i=1}^n X_i \right)^a$

Example: X_i : 2, 3, 4, 5

$Y_i: 1, 2, 3, 4$

$$1. \quad \prod_{i=1}^4 3 = 3^4 = (3)(3)(3)(3) = 81$$

$$2. \quad \prod_{i=1}^4 X_i = (2)(3)(4)(5) = 120$$

$$3. \quad \prod_{i=1}^4 5X_i = (5)^4 \prod_{i=1}^4 X_i = (5)^4 (120) = 75000$$

$$4. \quad \prod_{i=1}^4 Y_i = (1)(2)(3)(4) = 24$$

$$5. \quad \prod_{i=1}^4 X_i Y_i = \prod_{i=1}^4 X_i \prod_{i=1}^4 Y_i = (120)(24) = 2880$$

$$6. \quad \prod_{i=1}^4 \frac{X_i}{Y_i} = \frac{\prod_{i=1}^4 X_i}{\prod_{i=1}^4 Y_i} = \frac{120}{24}$$

$$7. \quad \prod_{i=1}^4 X_i^2 = \left(\prod_{i=1}^4 X_i \right)^2 = (120)^2 = 14400$$